

GRAAFF-REINET, South Africa

WMO: 687370

Lat: 32.193S

Lon: 24.543E

Elev: 2598

StdP: 13.37

Time Zone: 2.00 (E02)

Period: 98-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
6	28.7	31.4	6.9	8.6	66.6	12.2	11.3	61.4	21.1	63.3	19.0	64.6	2.8	10	0.418

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	24.6	96.9	63.9	93.4	63.4	90.4	63.1	70.6	82.2	68.9	80.9	67.6	79.9	10.4	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.3	110.5	75.6	65.6	104.0	73.5	63.9	97.8	72.7	36.2	82.0	34.7	81.0	33.6	79.7	77.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.7	16.9	15.0	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	64.3	74.8	74.6	70.4	64.0	59.0	53.3	55.0	54.9	60.2	65.2	68.6	72.3
	DBStd	10.32	5.95	5.37	5.98	7.46	7.98	7.34	8.73	8.35	7.18	7.55	6.44	5.63
	HDD50	148	0	0	0	2	12	46	47	36	4	1	0	0
	HDD65	1687	6	2	18	108	225	356	328	334	179	90	33	8
	CDD50	5365	769	688	633	421	290	144	201	188	312	471	558	690
	CDD65	1429	310	270	186	77	38	4	17	21	36	96	141	233
	CDH74	13889	3098	2345	1576	617	305	20	93	225	527	1165	1628	2290
	CDH80	5724	1463	1005	610	191	66	0	4	40	182	486	687	990
(14) Wind	WSAvg	6.6	7.1	6.6	6.2	5.4	5.8	6.6	7.0	6.1	6.3	7.1	7.3	7.7
(15) Precipitation	PrecAvg	12.0	1.2	1.5	1.9	1.1	0.7	0.5	0.6	0.7	0.4	1.0	1.1	1.3
	PrecMax	22.8	5.4	5.0	7.1	3.9	5.9	2.0	3.6	3.9	2.2	3.9	4.0	4.9
	PrecMin	5.7	0.0	0.0	0.0	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.0
	PrecStd	4.1	1.3	1.3	1.7	0.9	1.1	0.5	0.9	1.0	0.5	1.1	0.9	1.3
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	101.5	99.4	96.4	90.0	85.6	76.5	80.0	84.7	90.8	97.3	97.9	99.4
		MCWB	64.2	65.3	64.6	60.8	57.8	52.4	52.0	54.8	56.6	61.1	61.2	62.6
	2%	DB	96.1	93.9	91.0	85.3	81.2	73.7	76.6	79.9	85.6	91.1	91.9	93.9
		MCWB	64.8	66.0	64.7	60.7	56.9	51.2	51.7	51.7	56.1	60.1	61.0	62.6
	5%	DB	92.1	89.8	86.7	80.8	77.0	70.7	73.2	75.9	80.4	86.0	87.7	89.8
		MCWB	65.0	66.0	63.8	60.1	55.2	50.3	50.8	50.8	54.9	58.4	60.8	63.1
	10%	DB	88.1	86.1	82.7	76.4	72.9	66.8	69.7	71.6	75.3	80.9	83.6	85.8
		MCWB	64.6	65.6	63.1	59.0	53.7	49.0	49.5	49.7	53.7	58.1	60.1	63.1
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	71.9	73.5	71.8	65.5	60.5	54.7	55.4	59.6	61.7	65.4	68.2	71.0
		MCDB	86.1	85.1	82.8	76.3	77.7	70.8	72.5	73.2	79.7	84.8	83.0	81.4
	2%	WB	69.5	70.8	69.1	63.9	58.4	53.1	53.5	55.7	59.2	62.8	65.4	68.6
		MCDB	83.7	81.7	79.4	76.4	75.6	68.5	69.9	67.0	76.8	82.3	81.9	80.7
	5%	WB	67.9	69.1	67.2	62.3	56.9	51.6	52.1	53.6	57.3	61.3	63.6	67.1
		MCDB	80.8	80.4	78.2	74.6	72.5	66.3	68.9	66.5	74.4	78.5	80.1	79.4
	10%	WB	66.5	67.6	65.4	60.8	55.3	50.2	50.5	51.5	55.3	59.8	62.1	65.5
		MCDB	81.3	79.4	77.2	72.8	69.3	64.3	66.9	67.3	71.9	75.5	78.1	78.6
(35) Mean Daily Temperature Range	5% DB	MDBR	24.6	22.6	22.5	22.4	24.0	22.9	24.5	26.0	25.3	25.5	25.8	25.0
		MCDBR	31.8	29.1	29.5	27.5	27.8	26.0	25.8	29.4	34.7	34.2	34.2	32.1
		MCWBR	8.5	8.4	9.3	9.8	11.3	12.0	11.4	12.2	13.4	12.0	10.5	9.3
	5% WB	MCDBR	24.8	21.7	22.4	23.0	25.4	22.7	24.7	24.0	28.2	28.1	27.8	23.9
		MCWBR	8.4	8.5	9.1	9.2	12.2	12.2	12.4	11.6	12.6	10.7	10.0	8.6
(40) Clear-Sky Solar Irradiance	taub		0.346	0.353	0.338	0.309	0.288	0.274	0.274	0.318	0.342	0.358	0.334	0.340
	taud		2.496	2.476	2.496	2.561	2.598	2.605	2.593	2.445	2.361	2.361	2.477	2.499
	Ebn at Noon		315	306	300	293	285	282	287	285	293	301	317	318
	Edh at Noon		36	36	33	28	23	22	23	30	37	40	37	36
(44) All-Sky Solar Radiation	RadAvg		2430	2198	1843	1417	1112	983	1070	1342	1757	2112	2403	2578
	RadStd		118	112	108	58	42	54	58	75	105	96	122	146

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (6 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page